

SPARK 2026 | Breast Lesion Segmentation Challenge

Build a U-Net from scratch in PyTorch to segment breast tumours from ultrasound images

Overview

SPARK Academy 2026 | Breast Lesion Segmentation Challenge

Welcome to the **SPARK Academy 2026 | Breast Lesion Segmentation Challenge**.

Skilled radiologists are scarce across Sub-Saharan Africa, yet breast ultrasound scans are widely available. Your task is to build a model that segments breast lesions from ultrasound images — pixel by pixel — giving clinicians an automated tool for tumour localisation and measurement.

Goal: Build a breast lesion segmentation model from scratch in PyTorch. Start from U-Net, modify it, research improvements, and push your Dice score as high as you can.

Metric: Mean Dice Coefficient on the held-out test set.

Submission: A `.csv` file with `image_id` and `rle_mask` columns, plus a PDF summary and notebook emailed to `spark@cameramiafrica.org`. See the Rules tab for full details.

Description

The Problem

Breast cancer kills over 670,000 women annually. In Africa, radiologists are scarce — yet ultrasound machines are not. Your task: build an AI model that segments breast tumours from ultrasound images, pixel by pixel.

The Task

Binary semantic segmentation. Given a breast ultrasound image, predict a mask where **1 = lesion** and **0 = background**.

Scored using **mean Dice Coefficient**:

$$\text{Dice} = \frac{2 \times |P \cap G|}{|P| + |G|}$$

The Dataset

A curated combination of **BUSI** and **BUS-BRA** — two publicly available, expert-annotated breast ultrasound datasets. Deduplicated, standardised, and split into train, val, and test sets.

Split	Masks
Train	 Provided
Val	 Provided
Test	 Hidden

What You Must Do

- Implement **U-Net from scratch** in PyTorch as your baseline
- Modify the architecture, experiment with loss functions and augmentations
- Research improvements from the literature and apply them
- Document everything in your notebook

Pre-trained encoders and segmentation libraries are **not permitted**.

Submission Format

A `.csv` file with two columns: `image_id` (e.g. `sp_test_0001.png`) and `rle_mask` (run-length encoded prediction). Use `sample_submission.csv` as your template.

Dataset Citations

Al-Dhabyani et al. Dataset of breast ultrasound images. *Data in Brief*. 2020. <https://doi.org/10.1016/j.dib.2019.104863>

Gómez-Flores et al. BUS-BRA. *Medical Physics*. 2024. <https://doi.org/10.1002/mp.16812>

Evaluation

Submissions are evaluated on the **mean Dice Coefficient** between the predicted mask and the ground truth mask across all test images.

$$\text{Dice} = \frac{2 \times |P \cap G|}{|P| + |G|}$$

A score of **1.0** is a perfect prediction. A score of **0.0** means no overlap. The public leaderboard is scored on 50% of the test images. Final rankings are based on the remaining 50% revealed after the deadline.

Submission File

For each image in the test set, predict a run-length encoded binary mask. The file should contain a header and have the following format:

```
image_id,rle_mask
sp_test_0001.png,145 3 512 7 1024 2
sp_test_0002.png,
sp_test_0003.png,201 5 890 4
```

- `image_id` must match the test image filenames exactly
- `rle_mask` is the predicted mask encoded as `start length` pairs in column-major (Fortran) order
- If your model predicts no lesion for an image, leave `rle_mask` empty

Download `sample_submission.csv` from the Data tab to use as your template.